30-day All-Cause Hospital Readmission Prediction

STAT3612 Group 17

Name	UID	Work	Percentage
Huang Yasong	3036127892	Feature Engineering, Track 1 and Track 2 Model Coding Design, Presentation and Report (Discussion part)	25
Cui Yuxiang	3036218184	Track 1 Model Training, Hyper-Parameters, Experiment, Presentation and Report (Model and results)	25
Sun Weize	3036100363	Track 1 Model Training, Presentation and Report (Model Interpretation)	25
Shao Boning	3036127373	Track 1 Model Training, Multi Model Trial, Presentation and Report (Introduction and Data Processing)	25

1. Introduction

Preventable hospital readmissions represent a major global challenge for healthcare systems. Their incidence reflects both the overall health status of patients and the efficiency of resource utilization within hospitals.

With the widespread adoption of Electronic Health Records (EHR), machine learning offers unprecedented opportunities for predicting hospital readmission risk and can play a pivotal role in patient management and healthcare resource planning.

This report investigates the predictability of 30-day all-cause hospital readmissions using the publicly available, de-identified MIMIC-IV database. The study integrates structured EHR data—including demographics, laboratory tests, medications, and diagnoses—with unstructured clinical notes to obtain a comprehensive view of patient health. While imaging features were initially considered, they were excluded from the final modeling scope due to computational constraints. Nevertheless, the remaining multimodal data provide a rich and practical foundation for predictive analysis.

Preliminary exploration revealed a clear positive correlation between patient age and readmission rate, confirming that clinical features in the dataset contain meaningful predictive signals for patient outcomes. The primary objective of this study is to develop a robust binary classification model for readmission prediction, with performance evaluated using the Area Under the Receiver Operating Characteristic Curve (AUROC).

The remainder of this report is organized as follows: Section 2 introduces the data processing pipeline, including exploratory analysis and feature engineering. Section 3 presents model selection and performance comparison. Section 4 details the interpretation of the model and results, Section 5 further discusses limitation and potential good features, and Section 6 is a conclusion. Ultimately, this study aims to provide a transparent and practically applicable machine learning approach that supports data-driven improvements in patient management and healthcare quality.

2. Data Processing

To ensure the reliability of predictive modeling, we implemented a systematic data processing pipeline. This involved analyzing the characteristics of the raw data, addressing class imbalance, and applying targeted feature engineering strategies to convert complex time-series data into fixed-length feature vectors.

2.1 Dataset Overview and Exploratory Analysis

The dataset comprises approximately 82,465 admission records from 11,041 unique patients. The target variable is binary, indicating whether a patient was readmitted within 30 days of discharge. Exploratory Data Analysis (EDA) revealed a class imbalance, with a readmission rate of approximately 23%, necessitating careful handling during model training.

Structurally, the raw EHR data is three-dimensional (Samples × Time × Features). Each admission record functions as a "hospital diary," where rows represent the days of the hospital stay (Length of Stay), and columns represent various clinical indicators. The average length of stay is approximately 22.6 days, providing rich longitudinal data for analysis.

The 171 raw clinical features fall into four distinct categories:

- **Demographics:** Basic static profiles such as age and gender.
- **Laboratory Tests:** Dynamic tracks of vital signs and blood work results.
- **Medications:** Records of treatments administered during the stay.
- **Diagnoses (ICD Codes):** Representations of the patient’s disease history.

Before feature aggregation, we analyzed the volatility of these variables during patient stays. We observed a clear distinction in feature behavior:

- **Static Features:** Demographics, ICD Codes, and Medications were found to be almost entirely static, meaning these data points rarely change during a single hospital stay.
- **Dynamic Features:** In contrast, Laboratory Tests were mostly dynamic, fluctuating daily to reflect the patient's real-time physiological state.

2.2 Feature Engineering Strategy

Driven by the analysis of feature characteristics, we adopted a "Divide and Conquer" strategy to compress the time-series data into a tabular format suitable for machine learning models.

2.2.1 Processing Static Data

For features identified as stable (Demographics, ICD Codes, and Medications), computing averages would be redundant or meaningless. Therefore, we extracted the Last Value (the value recorded closest to discharge). This approach efficiently captures the patient's final status and confirms diagnoses upon leaving the hospital.

2.2.2 Processing Dynamic Data

For dynamic Laboratory Tests, capturing the trajectory of health changes is as important as the baseline values. Statistics of Mean, Std, Min, Max, Slope, Delta are calculated. Delta is the difference between the values of the first day and last day of one admission. In addition to standard statistics, we introduced a specialized interaction term: Slope-times-Days. The Slope (calculated via least squares regression) represents the rate of change in a vital sign. Days represent the duration of the stay. Multiplying these reveals the total magnitude of health change during hospitalization. This feature is more robust and interpretable than simple differences, as it accounts for the trend over the entire duration of the stay.

2.2.3 Processing Clinical Notes

To incorporate unstructured text data, we applied TF-IDF (Term Frequency-Inverse Document Frequency) Vectorization. This technique converted clinical notes into numerical features, allowing the model to capture semantic risks and doctor observations that are not reflected in structured tabular data.

Following these processing steps, the final dataset resulted in a feature dimension of 972, providing a comprehensive input vector for the subsequent modeling phases.

3. Model and Result

3.1 Overview

In this section, we present the modeling approach and experimental results. We used three state-of-the-art gradient boosting frameworks—XGBoost, LightGBM, and CatBoost—to ensure a robust prediction capability. To further enhance performance and reduce variance, we implemented a Model Fusion strategy using Rank Averaging.

3.2 Methodology

For all models, we adhered to a rigorous training and validation method:

- **Cross-Validation:** We utilized a 5-Fold Stratified Cross-Validation strategy in the training set to ensure that the class distribution remained consistent across all folds, providing a reliable estimate of model performance. And good for small datasets.
- **Optimization:** The Optuna library was employed for hyper-parameter tuning (trial.suggest_float) to identify the optimal combination for each model.
- **Class Imbalance:** To address data imbalance, parameters of neg_pos_ratio were used accordingly.
- **Evaluation Metric:** The Area Under the Curve (AUC) was used as the primary metric for evaluation.
- **Validation**: Using Validation set to test whether the model is overfitted. Experiments show that combining training and validation sets for cross validation does not outperform using only the training set.

3.3 Model 1: XGBoost (Extreme Gradient Boosting)

XGBoost was chosen for its scalable tree boosting system and regularization capabilities. We configured the model to prioritize a slow learning rate combined with a high number of boosting rounds to reach the global minimum of the loss function without overfitting.

3.3.1 Hyperparameter Configuration

The optimization process yielded a specific set of parameters designed to balance model complexity and regularization. Notably, a relatively low learning rate (eta ≈ 0.01) was paired with significant regularization (gamma, lambda, alpha) to control variance.

Parameter Category	**Parameter Name**	**Optimized Value**	**Description**
Tree Structure	max_depth	7	Maximum depth of a tree.
	min_child_weight	3.60872	Minimum sum of instance weight needed in a child.
	max_delta_step	1.70060	Constraint for weight update (useful for imbalance).
Learning	eta (Learning Rate)	0.01053	Step size shrinkage used to

Process			prevent overfitting.
	num_boost_round	4000	Maximum number of boosting iterations.
	early_stopping_rounds	200	Stop training if the validation score doesn't improve.
Stochastic Sampling	subsample	0.90490	Subsample ratio of the training instances.
	colsample_bytree	0.93318	Subsample ratio of columns when constructing each tree.
	colsample_bylevel	0.96201	Subsample ratio of columns for each level.
Regularization	gamma	2.64623	Minimum loss reduction required to make a further partition.
	lambda (L2)	0.65644	L2 regularization term on weights.
	alpha (L1)	0.01594	L1 regularization term on weights.
Imbalance	scale_pos_weight	1.48059	Balancing of positive and negative weights.
Misc	seed	4	Random number seed.

3.3.2 Results

- **Training:** The model fitted the training data with a Train AUC of **0.98849**.
- **Validation:** The 5-fold validation AUC was **0.788981**.
- **Test (Kaggle):** Achieving a score of **0.82**.

3.4 Model 2: LightGBM (Light Gradient Boosting Machine)

LightGBM was utilized for its efficiency and leaf-wise tree growth strategy, which often results in faster training and higher accuracy on large datasets.

3.4.1 Hyperparameter Configuration

A key difference in the LightGBM configuration was the allowance for complex trees (num_leaves: 161) constrained by max_depth. The model also utilized a higher scale_pos_weight (2.45) compared to XGBoost, indicating it required stronger weighting to handle the class imbalance.

Parameter Category	Parameter Name	Optimized Value	Description
Tree Structure	num_leaves	161	Maximum tree leaves for base learners.
	max_depth	11	Maximum tree depth.
	min_child_samples	90	Minimum number of data needed in one leaf.
Learning Process	learning_rate	0.01054	Learning rate.
	num_boost_round	4000	Total boosting iterations.
Sampling	subsample	0.89702	Bagging fraction.
	subsample_freq	4	Frequency for bagging.
	colsample_bytree	0.67820	Feature fraction.
Regularization	reg_alpha	0.00183	L1 regularization.
	reg_lambda	0.00336	L2 regularization.
Imbalance	scale_pos_weight	2.45842	Weight for the positive class.

3.4.2 Results

- **Training:** Train AUC reached **0.960436**.

- **Validation:** Validation AUC was **0.781387**.
- **Test (Kaggle):** The model achieved a score of **0.82**.

3.5 Model 3: CatBoost (Categorical Boosting)

CatBoost was selected specifically for its capability to handle categorical features automatically and its symmetric tree structure, which often reduces overfitting.

3.5.1 Hyperparameter Configuration

The CatBoost model required the highest number of boosting rounds (5000) and the strongest imbalance correction (scale_pos_weight: 4.69). It also employed Bayesian bootstrapping and significant L2 leaf regularization to stabilize the learning process.

Parameter Category	**Parameter Name**	**Optimized Value**	**Description**
Tree Structure	depth	9	Depth of the tree.
Learning Process	learning_rate	0.01773	Learning rate.
	num_boost_round	5000	Total boosting iterations.
Regularization	l2_leaf_reg	9.74124	Coefficient at the L2 regularization term of the cost function.
	random_strength	0.02525	The amount of randomness for scoring splits.
	bagging_temperature	0.51221	Controls the intensity of Bayesian bagging.
Imbalance	scale_pos_weight	4.69231	Weight for the positive class.
Sampling	bootstrap_type	Bayesian	Bayesian bootstrapping method.

3.5.2 Results

- **Training:** Train AUC was **0.923303**.
- **Validation:** Validation AUC was **0.786633**.
- **Test (Kaggle):** The test score was **0.79**.

3.6 Model Fusion: Rank Averaging

While individual models demonstrated strong performance, ensemble methods often yield better generalization by mitigating the biases of single algorithms. We implemented a **Rank Averaging** fusion.

3.6.1 Methodology

Probability outputs from different models can have different distributions (e.g., one model migh t cluster predictions around 0.1 and 0.9, while another clusters around 0.4 and 0.6). Averaging raw probabilities can be skewed by these distribution differences.

Rank averaging resolves this by converting raw probabilities into ranks (normalized between 0 and 1) and then averaging the ranks.

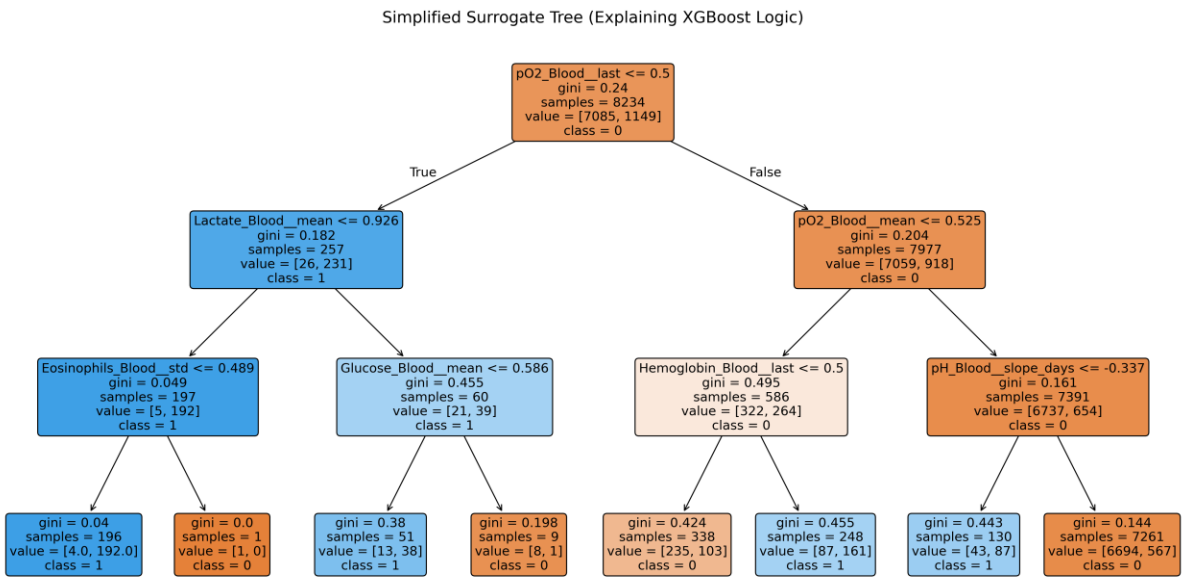
$$\text{Rank}_{\text{Final}} = \frac{\text{Rank(XGB)} + \text{Rank(LGB)} + \text{Rank(Cat)}}{3}$$

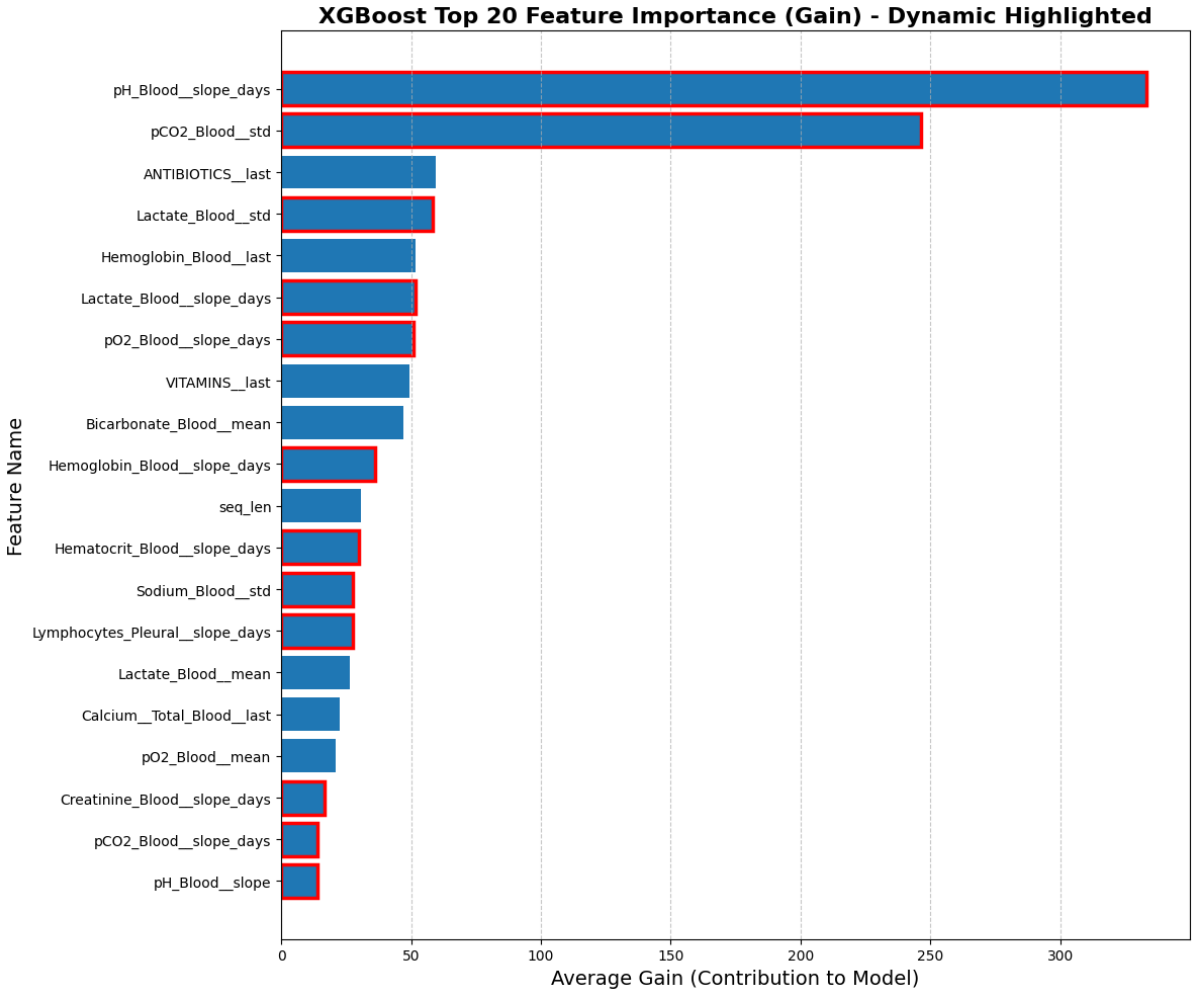
3.6.2 Final Performance

The fusion model successfully combined the strengths of XGBoost, LightGBM, and CatBoost.

- **Validation (Train + Val set):** The ensemble achieved an improved AUC of **0.814595** and an AUPRC of **0.646098**.
- **Test (Kaggle):** The final submission achieved a score of **0.82**, in order to enhance the robustness, this model has been chosen as the final model.

4. Model Analysis





The XGBoost feature-importance analysis for 30-day readmission shows that the model relies mainly on trends over time—slopes and variability of key laboratory values—rather than single measurements at discharge. This pattern is clinically plausible: dynamic changes in physiology (e.g., whether a parameter is improving or drifting in the wrong direction) often predict short-term deterioration better than one "normal" value at a single time point. Similar findings have been reported in sepsis and critical-care research, where trajectories such as lactate clearance outperform isolated values for prognostication (Espinosa & Brown, 2021; Sugimoto et al., 2021; Zhang & Xu, 2014).

4.1 Physiological Stability (Acid-Base & Respiratory Trends)

The most influential features in the model relate to pH and pCO_2 dynamics (e.g., pH_Blood__slope_days, pCO2_Blood__std). A normal pH at discharge may still be unsafe if it has been trending rapidly toward acidosis or alkalosis, because this pattern suggests failing compensatory mechanisms and unresolved disease. In parallel, large fluctuations or upward trends in pCO_2 indicate unstable ventilation and possible chronic hypercapnia, which is associated with increased morbidity and mortality in COPD and other respiratory diseases (Csoma et al., 2022). These acid–base and respiratory features therefore act as markers of physiological instability, providing a mechanistic explanation for their strong association with early readmission risk.

4.2 Hidden Metabolic Stress (Lactate Clearance)

Features capturing lactate behavior over time (e.g., Lactate_Blood__slope_days,

Lactate_Blood__std, Lactate_Blood__mean) reflect tissue perfusion and global metabolic stress. Elevated lactate and poor lactate clearance are consistently linked to higher mortality and worse outcomes in sepsis and critical illness, and lactate-guided strategies have been recommended in major guidelines (Espinosa & Brown, 2021; Sugimoto et al., 2021; Zhang & Xu, 2014). In this context, a lack of sustained lactate decline or highly variable lactate values near discharge plausibly indicate incomplete resolution of shock, infection, or low cardiac output, which in turn increases the likelihood of decompensation and readmission within 30 days.

4.3 Indicators of Frailty and Unresolved Infection

Medication features such as **(ANTIBIOTICS__last) and (VITAMINS__last)** behave as clinical proxies rather than causal factors. Ongoing or discharge antibiotics usually reflect recent or partially treated serious infections, and infectious diagnoses such as sepsis are among the leading causes of 30-day readmission in national datasets. Likewise, continued vitamin supplementation in an acute-care setting often marks patients who are malnourished, alcohol-dependent, or generally frail. Frailty itself is a well-established predictor of unplanned readmission in older adults (Gregersen et al., 2020; Pedersen et al., 2017). It is therefore consistent that the model treats these medication “last value” indicators as high-value signals of underlying vulnerability and infection-related risk.

4.4 Oxygen Capacity Risks (Anemia Trends)

Hemoglobin and hematocrit features (e.g., Hemoglobin_Blood__last, Hemoglobin_Blood__slope_days) capture both baseline oxygen-carrying capacity and ongoing blood loss or bone-marrow failure. Anemia is common in patients with heart failure and has been associated with higher hospitalization and mortality rates, and in some cohorts with 30-day readmission risk (Gangu et al., 2022; Zhang et al., 2014). A downward hemoglobin trend near discharge may indicate occult bleeding or progressive disease, even if the absolute level has not yet become critically low. Patients with limited cardiopulmonary reserve and actively declining hemoglobin are therefore more likely to become symptomatic—developing dyspnea, syncope, or heart-failure exacerbation—and to re-present to hospital within 30 days.

5. Further Discussion

In this section, we conducted further discussion on potential feature design that could be useful in real-world scenarios. Our dataset exhibits certain limitations when compared to real clinical data — most notably, the absence of hospitalization time information. Despite this, we observed that some patients, identified by their unique subject IDs, have multiple hospitalization records.

This observation led us to hypothesize that a patient’s *historical hospitalization records* might provide valuable information and could be utilized as a new feature to enhance prediction performance. To explore this idea, we conducted a preliminary experiment by adding one additional feature, Patient_Visit_Count, which represents the number of times a patient appears in the entire dataset. The same model architecture was then retrained using this augmented feature set.

The results showed a notable improvement in model performance, with the AUC increasing from 0.82 to 0.89. However, because the original dataset lacks temporal information, this

feature technically introduces data leakage, as it unintentionally incorporates information from both past and future visits relative to the prediction target.

Although this finding is not strictly valid, it nonetheless indicates that historical hospitalization records can serve as valuable predictors in real-world applications. Intuitively, such data reflect the patient's previous health conditions and therefore provide critical context for predicting future outcomes. Consequently, in realistic clinical settings where time-stamped hospitalization data are available, these historical features could constitute an important and effective source of predictive information.

6. Conclusion

This study developed and evaluated machine learning models for predicting 30-day all-cause hospital readmission using structured EHR data and unstructured clinical notes from the MIMIC-IV database. Through systematic feature engineering and extensive model comparison, gradient boosting methods—particularly XGBoost and LightGBM—demonstrated strong predictive capability, achieving an AUROC of approximately 0.82 on the test set.

Feature-importance analysis revealed that temporal trends in laboratory indicators, rather than single-point measurements, were the most informative predictors of readmission, reflecting the clinical importance of physiological stability and disease trajectory. Furthermore, exploratory investigation into patient historical visit information suggested that incorporating longitudinal admission data could further enhance model performance in realistic clinical settings.

Overall, this work confirms the feasibility of data-driven readmission risk prediction and highlights the value of integrating structured and unstructured EHR data. Future research should focus on incorporating temporal hospitalization context and external validation across diverse hospital systems to enhance both generalizability and clinical applicability.

References

Csoma, B., Vulpi, M. R., Dragonieri, S., Bentley, A., Felton, T., Lázár, Z., & Bikov, A. (2022). Hypercapnia in COPD: Causes, consequences, and therapy. *Journal of Clinical Medicine, 11*(11), 3180.

Espinosa, K., & Brown, S. R. (2021). Serum lactate testing to predict mortality in patients with sepsis. *American Family Physician, 103*(5), 309–310.

Gangu, K., Bobba, A., Patel, P. S., Vellanki, S., Trikannad, A. K., Basida, S., Avula, S., & Tripathi, A. (2022). In-hospital mortality rate and predictors of 30-day readmission in patients with iron deficiency anemia and diastolic heart failure: A cross-sectional study. *American Journal of Hospital Medicine, 6*(3).

Gregersen, M., Hansen, T. K., Jørgensen, B. B., & Damsgaard, E. M. (2020). Frailty is associated with hospital readmission in geriatric patients: A prognostic study. *European Geriatric Medicine, 11*, 783–792.

Sugimoto, M., Takayama, W., Murata, K., & Otomo, Y. (2021). The impact of lactate clearance on outcomes according to infection sites in patients with sepsis: A retrospective observational study. *Scientific Reports, 11*, 22394.

Zhang, Z., & Xu, X. (2014). Lactate clearance is a useful biomarker for the prediction of all-cause mortality in critically ill patients: A systematic review and meta-analysis. *Critical Care Medicine, 42*(9), 2118–2125.